TEXORA · LECTURE SHEET

Production Calculation: Line Balancing, Floor Optimization, Critical Path, ERP and MRP

Four planning calculations of the apparel factory: balancing operators against the pitch, scheduling an order on its critical path, and exploding the bill of materials into purchase requirements through MRP inside an ERP system.

AM-3207 · ADVANCED MACHINERIES AND RELEVANT CALCULATIONS · TOPIC 8 · SPECIALIZATION — APPAREL MANUFACTURING

1. Line balancing

A sewing line runs at the speed of its slowest station. **Line balancing** distributes operators across operations so that every station's time per piece is as equal as possible — and no station exceeds the **pitch time** set by the target. The procedure: list operation SMVs; fix the required output, hence the pitch; assign operators per operation = SMV ÷ pitch, rounded up; the station with the highest resulting time per piece is the **bottleneck** and sets the line's real output.

WORKED EXAMPLE — BALANCING A 5-OPERATION SECTION

Target 60 pcs/h → pitch = 60 ÷ 60 = **1.00 min/piece**. Operations (SMV → operators → time/piece): collar make 2.0 → 2 → 1.00; collar attach 3.0 → 3 → 1.00; sleeve attach 2.5 → 3 → 0.83; side seam 1.5 → 2 → 0.75; hemming 1.0 → 1 → 1.00. Total SMV = 10.0 min; total operators = **11**.

Bottleneck = 1.00 min → line output = 60 ÷ 1.00 = **60 pcs/h = 480 pcs per 8-h shift**.

Balance efficiency = (480 × 10.0) ÷ (11 × 480) × 100 = **90.9%**. The 9.1% loss is the rounding tax: sleeve attach runs at 0.83 and side seam at 0.75 because operators cannot be divided — remedies are combining operations, multi-skill operators who float between the two light stations, or a work aid that cuts the bottleneck SMV.



Figure 1 — Operator assignment per operation (bars to scale). Stations at 1.00 min/pc are balanced to the pitch; the two below it carry the unavoidable rounding loss.

Floor optimization is the physical side of the same calculation: stations are laid out in operation sequence (straight or U-shaped, the U bringing the line's start and finish — and its supervisor — together), spaced at the machine-plus-operator footprint the balance assumes, with bundle or hanger transfer distances minimised. A balanced line drawn on a bad floor plan still loses its minutes to walking and waiting.

2. Critical path method (CPM) for an order

An export order is a project: bookings, approvals, production and shipment on a deadline. **CPM** networks the activities with their durations and dependencies; the longest chain — the **critical path** — fixes the earliest completion, and every other activity carries **float** (slack) equal to how far it sits off that chain.

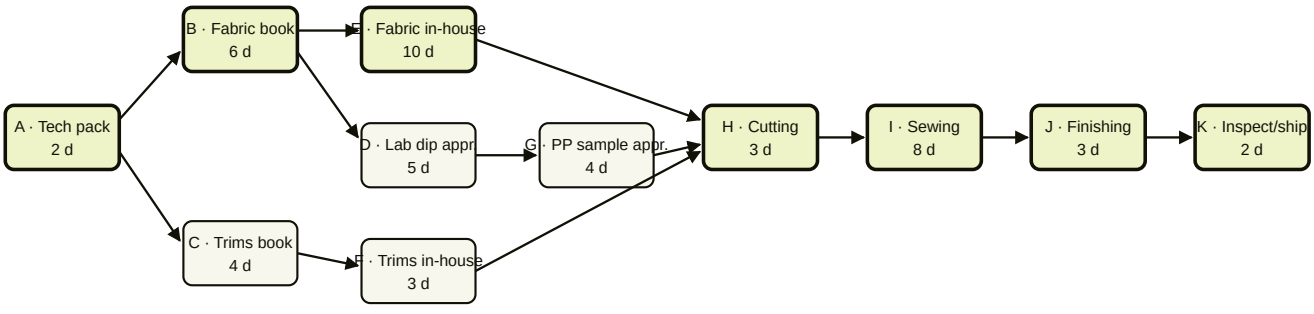


Figure 2 — CPM network for the course order. Durations in working days; arrows are dependencies (an activity starts when all its incoming activities finish).

ACTIVITY	DURATION (D)	EARLIEST START–FINISH	FLOAT (D)
A Tech pack & order confirmation	2	0–2	0
B Fabric booking	6	2–8	0

C Trims booking	4	2–6	9
D Lab-dip approval	5	8–13	1
E Fabric in-house	10	8–18	0
F Trims in-house	3	6–9	9
G PP sample approval	4	13–17	1
H Bulk cutting (needs E, F, G)	3	18–21	0
I Bulk sewing	8	21–29	0
J Finishing & packing	3	29–32	0
K Final inspection & shipment	2	32–34	0

The three chains confirm the network: A–B–E–H–I–J–K = 2+6+10+3+8+3+2 = **34 days**; through lab-dip and PP sample = 33 days (float 1); through trims = 25 days (float 9). Management attention belongs on the critical chain: a two-day delay in fabric in-house moves the shipment two days, while trims could slip a week without consequence. If the buyer's deadline is day 32, the plan must *crash* the critical path — faster fabric (air-freighted greige, a mill with shorter lead time) or overlapped sewing — never by exhorting the trims department.

3. ERP and MRP

ERP (enterprise resource planning) software holds the factory's single shared database — orders, BOMs, inventory, production, HR and accounts — so every department calculates from the same numbers. **MRP** (material requirements planning) is the calculation engine inside it: for each order it **explodes** the bill of materials (Topics 4 and 5) into gross requirements, subtracts stock, and issues net purchase requirements with need dates taken from the CPM schedule.

WORKED EXAMPLE — MRP NETTING FOR THE COURSE ORDER

ITEM	GROSS REQUIREMENT	IN STOCK	NET TO PURCHASE
Fabric	11,124 m	1,124 m	**10,000 m**
Buttons	123,600 pcs	3,600 pcs	**120,000 pcs**
Sewing thread (main shade)	37 cones	5 cones	**32 cones**

Gross requirements come straight from Topics 4 and 5; MRP's contribution is the netting and the timing — fabric must be ordered at day 2 (activity B) to be in-house by day 18, while buttons, on

> the 9-day-float trims chain, are safely ordered later.

Glossary

TERM	MEANING
Bottleneck	The station with the highest time per piece; it fixes the line's output.
Balance efficiency	Standard minutes earned by the line's output ÷ total operator-minutes attended.
Critical path	The longest dependency chain through a project network; it sets the earliest completion date.
Float	Time an activity can slip without delaying the project.
MRP	Material requirements planning: exploding the BOM into gross requirements, netting stock, and scheduling purchases.
ERP	Enterprise resource planning: the integrated software holding one shared database for all departments.

Key points and summary

- Balancing assigns operators = SMV ÷ pitch; the bottleneck sets output; rounding operators costs balance efficiency (worked: 90.9%).
- CPM: the critical path A–B–E–H–I–J–K fixes the order at 34 days; lab-dip chain floats 1 day, trims 9 days.
- Only critical activities move the shipment date — crash those, ignore the float-rich ones.
- MRP turns Topics 4–5 calculations into net purchase orders (fabric 10,000 m, buttons 120,000, thread 32 cones), timed by the CPM plan inside the ERP.

Practice

Answer in your own words; brief model answers follow.

1. An operation of SMV 4.5 min must meet a pitch of 1.5 min. How many operators are assigned, and what is the station's time per piece?
2. In the CPM network, fabric in-house (E) is delayed by 3 days and trims in-house (F) by 5 days. What is the new project duration, and why?
3. MRP shows a gross requirement of 50,000 m of a fabric, stock of 8,000 m, and 6,000 m already on order. What is the net requirement?

4. Why is exhorting the trims department pointless when the shipment is late on this network?

Answer key

1. 4.5 ÷ 1.5 = 3 operators; time per piece = 4.5 ÷ 3 = 1.5 min — exactly at pitch.
2. E is critical: +3 days moves H and everything after it → 37 days. F has 9 days of float: +5 is absorbed, no effect. New duration 37 days.
3. 50,000 − 8,000 − 6,000 = 36,000 m.
4. The trims chain carries 9 days of float and does not touch the critical path; the shipment date is set by fabric, cutting and sewing. Effort must go to critical activities.

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